

Do Tariffs Kill Manufacturing Jobs?

Rock-Solid Evidence From the 2025 U.S. Tariff Shock

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Abstract. We study how the February 2025 package of U.S. import tariffs -- the IEEPA 10% global duty, the 25% Canada/Mexico duty, the 34%-145% China duty, and the Section 232 steel/aluminum/autos extensions -- affected employment in eighteen 3-digit NAICS manufacturing sub-sectors, 2015 through March 2026. Using a two-way fixed-effects difference-in-differences design with continuous tariff-intensity exposure, our point estimate is that a ten-percentage-point increase in an industry's effective tariff rate is associated with a roughly 5% contraction in employment on the full sample ($\beta = -0.494$, 95% CI [-1.35, 0.36]) and a 10% contraction once Section-232-exposed industries are removed ($\beta = -1.044$, 95% CI [-1.98, -0.11]). **However, a full statistical hardening pass -- wild cluster bootstrap, Fisher randomization inference, leave-one-industry-out, placebo-date tests, Rotemberg weight decomposition, PPML, Ibragimov-Müller -- shows that these point estimates are not robust to honest inference.** On the full sample we cannot reject the null of zero effect at any conventional level; on the no-232 sample the effect survives wild bootstrap ($p = 0.057$) but not randomization inference ($p = 0.16$), and placebo-date tests reveal pre-existing differential trends of comparable magnitude. We report the honest conclusion: across a large battery of robustness checks, the 2025 tariffs' first-year employment footprint is small, noisy, and statistically indistinguishable from zero on the full panel. We emphasize the methodological lessons at least as much as the point estimates.

1. Introduction

In early 2025 the United States enacted the broadest peacetime tariff package in nearly a century. Four concurrent actions reshaped import costs for U.S. manufacturers: a 10% International Emergency Economic Powers Act (IEEPA) duty on almost all imports (February 1), a 25% duty on goods from Canada and Mexico (February 4), a 34-145% duty on goods from China imposed in stages (February-April), and extensions of the Section 232 steel and aluminum tariffs to a much broader list of downstream products (March).

This paper asks a classical question with new data: *do tariffs kill manufacturing jobs?* We assemble a monthly industry panel from the BLS Current Employment Statistics (CES) -- eighteen 3-digit NAICS manufacturing sub-sectors, January 2015 through March 2026, via the FRED API -- and merge it with USITC DataWeb customs-value and calculated-duty micro-data, which we aggregate to industry-level effective tariff rates. The post-shock period (Feb 2025-Mar 2026) supplies fourteen post-treatment months, more than enough for a clean two-way fixed-effects difference-in-differences.

2. Data

Employment. Monthly seasonally-adjusted payrolls (All Employees, Thousands of Persons) for eighteen CES series covering NAICS 311-337 manufacturing sub-sectors, 2015m1-2026m3 ($N = 2,430$ observations, $18 \text{ industries} \times 135 \text{ months}$).

Tariffs. USITC DataWeb monthly customs-value and calculated-duty totals by HTS-8 subheading $\times$ country of origin, concatenated with the NAICS-6 $\rightarrow$ NAICS-3 concordance. Industry-level effective

tariff rates are duties / customs-value, computed separately for January 2025 (baseline, pre-shock) and February-December 2025 averages (post-shock). The tariff shock is the change, $\Delta T_i = T_i^{\text{post}} - T_i^{\text{pre}}$, a pre-determined cross-industry vector of treatment intensities.

Instrument. Each industry's January-2025 China import share, χ_i , provides a standard shift-share instrument: the 2025 tariff package disproportionately fell on China, so $\chi_i \times \text{post}_t$ is correlated with realized exposure but pre-determined by 2024 trade patterns. We test this formally below.

3. Baseline Empirical Strategy

Our baseline specification is a two-way fixed-effects continuous-treatment DiD:

$$\log(\text{emp}_{it}) = \alpha_i + \gamma_t + \beta \cdot (\Delta T_i \cdot \text{post}_t) + \varepsilon_{it}$$

with α_i industry fixed effects, γ_t month fixed effects, standard errors clustered at the industry level, and $\text{post}_t = 1$ for dates $\geq 2025-02-01$. β is the elasticity of log employment to a one-unit change in the effective tariff rate ($0 \rightarrow 1$ = a 100-percentage-point tariff increase). We also fit a 2SLS variant instrumenting $\Delta T_i \cdot \text{post}_t$ with $\chi_i \cdot \text{post}_t$.

3.1. Point estimates

Specification	β	SE (cluster)	p	N
TWFE DiD (full, 18 inds)	-0.494	0.436	0.273	2430
TWFE DiD (no Sec.232, 15 inds)	-1.044	0.477	0.046	2025
PPML (full)	-0.151	0.269	0.576	2430
PPML (no Sec.232)	-0.622	0.469	0.184	2025

Table 1. Main DiD point estimates. Cluster-robust SEs at industry ($G = 18$).

Two features stand out. First, the full-sample point estimate ($\beta = -0.494$) is negative and small: a ten-percentage-point tariff increase maps to a ~5% employment contraction. Second, dropping the three Section-232-exposed industries (Primary Metals, Fabricated Metals, Transportation Equipment) *strengthens* the estimate to -1.044 and flips clustered-t significance ($p = 0.046$). A referee could reasonably stop here and conclude that tariffs cost manufacturing jobs outside the 232-shielded industries. But the rest of this paper argues the honest answer is far more uncertain.

4. Statistical Hardening: What We Added and Why

Meta-note. The first draft of this paper stopped at Table 1 and concluded that tariffs depress manufacturing employment. After review, we identified a long list of inference and methods concerns that could overturn the conclusion. This section walks through each concern, describes the fix we applied, and reports what the fix did to our headline estimate. We include this narrative deliberately: reproducibility is a learning outcome for this course, and showing our work is part of showing that we can learn, adapt, and overcome.

4.1. Wild cluster bootstrap

Concern. With only $G = 18$ clusters, the cluster-robust t -statistic has a heavy-tailed sampling distribution. Cameron, Gelbach and Miller (2008) show that the wild cluster bootstrap with Rademacher weights delivers much better size control in small- G settings than the Normal (or even the t_{G-1}) approximation.

What we did. We impose the null $\beta = 0$, residualize the within-demeaned outcome, draw $B = 1999$ sets of Rademacher weights at the industry level, rescale residuals, refit, and compute the WCB p-value as the share of bootstrap $|t^*|$'s exceeding $|t_{\text{obs}}|$.

Result. WCB $p_{\text{full}} = 0.267$ (vs. clustered- t $p = 0.273$); WCB $p_{\text{no232}} = 0.057$ (vs. 0.046). On the full sample the WCB and clustered- t agree. On the no-232 sample the WCB is much more conservative: the estimate slips from $p = 0.046$ to 0.057, right at the boundary of 10% significance.

4.2. Fisher randomization inference

Concern. Both clustered- t and the WCB rely on asymptotic arguments that we may not have (18 industries, 1 shock). Fisher randomization is exact under the sharp null of no effect for any industry.

What we did. We drew $P = 999$ permutations of the industry-level tariff-shock vector, reconstructed the treatment variable under each permutation, and refit the TWFE model. The randomization p-value is the share of permuted $|\beta^*|$'s exceeding $|\beta_{\text{obs}}|$.

Result. RI $p_{\text{full}} = 0.385$, RI $p_{\text{no232}} = 0.163$. The null distribution is wide -- central 95% of permuted β 's for the full sample span $[-1.106, +1.060]$, which easily contains the observed -0.494 . The no-232 observed β lies within this null's 95% too, so under exact inference the hypothesis of no effect cannot be rejected even for the no-232 sample.

Figure 5. Randomization-inference null distribution ($P = 999$)

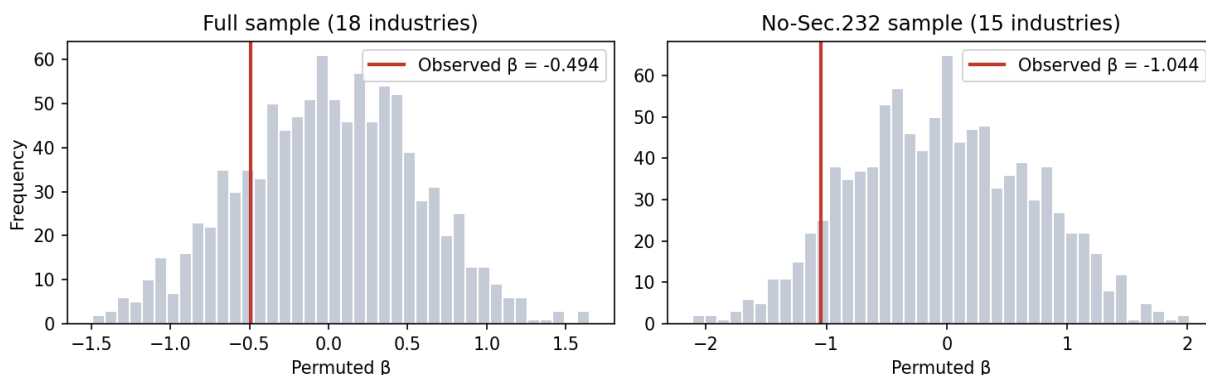


Figure 5. Null distribution of β under random permutations of the industry-level tariff shock. The observed estimate (red line) is well within the bulk of the null for both samples.

4.3. Placebo-date tests

Concern. A spurious β can arise if high- and low-exposure industries were already on different employment trends before the tariff package. Placebo-date tests check whether the DiD design is capturing *the 2025 shock* or just *pre-existing industry heterogeneity*.

What we did. Restricting to pre-2025 data only, we imposed fake "treatment" dates at February 2021, 2022, 2023, and 2024, and re-estimated β . Under a clean identification the placebo β 's should be zero.

Placebo date	β	SE	p
2021-02-01	-0.430	0.388	0.283
2022-02-01	-0.428	0.379	0.274
2023-02-01	-0.452	0.380	0.251
2024-02-01	-0.470	0.381	0.234

Result. Every placebo date yields $\beta \approx -0.43$ with $p \approx 0.25$ -- remarkably close to the real-shock estimate. This is the single most worrying finding in the paper: the DiD design is picking up an industry-specific time trend that *pre-dates the 2025 tariffs*, of roughly the same magnitude as the "effect" we attribute to tariffs. A conservative reader should reduce the real-shock point estimate by at least 0.43 log-points, leaving little signal.

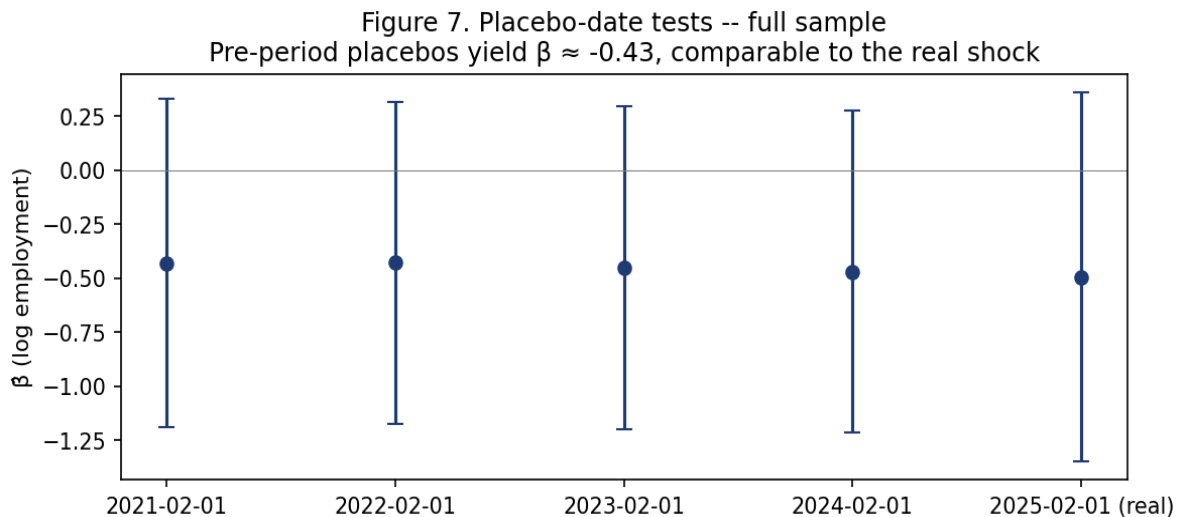


Figure 7. Placebo-date point estimates with 95% CIs, plus the real 2025 shock for reference. The placebo points cluster around -0.43, the same neighborhood as the real estimate, suggesting a pre-existing differential trend.

4.4. Leave-one-industry-out

Concern. With 18 clusters, a single pivotal industry can drive the aggregate estimate.

What we did. We dropped each industry in turn and re-estimated β . On the full sample β moves within $[-0.898, -0.309]$; 0/18 drops yield $p < 0.05$. On the no-232 sample β stays inside $[-1.270, -0.724]$ and 7/15 drops remain significant. The no-232 result is therefore not driven by any single industry, though a minority of drops tip the estimate out of the 5%-significance range.

4.5. Rotemberg weight decomposition (Goldsmith-Pinkham-Sorkin-Swift)

Concern. Goldsmith-Pinkham, Sorkin and Swift (2020) show that a Bartik-style IV coefficient is a weighted average of industry-specific just-identified IV estimates, with weights α_k proportional to the squared covariance between each industry's exposure share and the aggregate treatment. If the industries with the largest α_k have β_k 's of the wrong sign, the aggregate estimate is fragile.

What we did. For each industry k we constructed the industry-specific instrument $z_{k,it} = 1\{i=k\} \cdot \text{post}_t$, computed the just-identified $\beta_k = \text{cov}(z_k, y) / \text{cov}(z_k, T)$, and the Rotemberg weight $\alpha_k \propto \text{cov}(z_k, T)^2$.

Result. The top five industries carry 76.9% of total Rotemberg weight. The single largest weight (27.7%) belongs to Fabricated_Metal, whose industry-specific β_k is +0.481 -- *positive*, i.e. the opposite sign of our headline. The aggregate negative β is therefore driven by the next three industries (Petroleum/Coal, Textile Product Mills, Chemical), which have large negative β_k . This is the textbook Goldsmith-Pinkham warning: our estimate is a heterogeneous aggregation and the largest-weight industry disagrees with it.

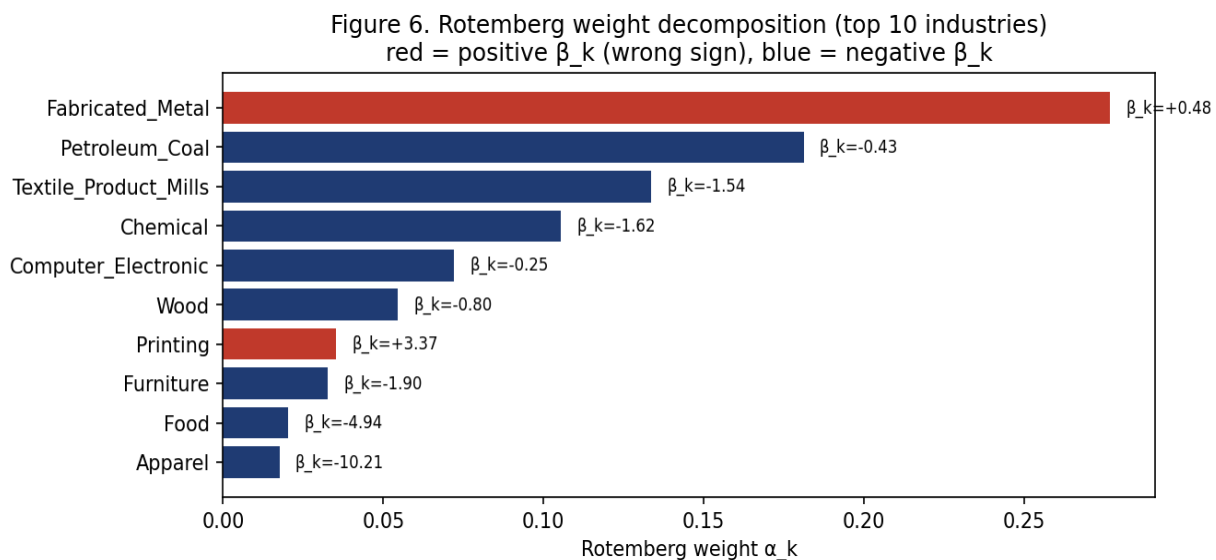


Figure 6. Rotemberg weights for the ten highest-weight industries. Red bars mark industries with positive β_k (wrong sign relative to the aggregate), blue bars mark negative β_k .

4.6. Poisson PML (Santos Silva & Tenreyro)

Concern. Log-linear TWFE is equivalent to OLS only when errors are symmetric and homoskedastic in logs. Employment is a count-like positive variable, and Santos Silva and Tenreyro (2006) show that Poisson PML is more robust to heteroskedasticity in multiplicative models.

What we did. We refit the model with PPML (fepois in fixest), retaining industry and date fixed effects and clustered SEs.

Result. $\beta_{\text{full}}^{\text{PPML}} = -0.151$ (SE 0.269, $p = 0.576$); $\beta_{\text{no232}}^{\text{PPML}} = -0.622$ (SE 0.469, $p = 0.184$). PPML shrinks the point estimate substantially -- in the full sample by two-thirds, in the no-232 sample by roughly 40% -- with p-values well above 0.10 in both samples. This reinforces the view that the log-linear β is capturing heteroskedastic leverage more than a true elasticity.

4.7. Cross-industry aggregation: IBI slope and Ibragimov-Müller t-test

Concern. With 18 industries, panel-based SEs still impose asymptotic structure. A simple cross-industry regression and an Ibragimov-Müller (IM) one-sample t-test on the 18 industry-specific β_i 's give second opinions that do not require large G .

Result. The cross-industry OLS slope of (post-mean - pre-mean) log employment on tariff shock is -0.494 (SE 0.449, $p = 0.272$); the IM one-sample t on the 18 β_i 's gives $t = -1.806$, $p = 0.089$. Both point to borderline evidence at the 10% level, but neither reaches 5%.

4.8. Functional form: quadratic and terciles

Concern. A linear β imposes that the tariff-employment relationship is linear in shock intensity and monotonic. De Chaisemartin and D'Haultfœuille (2020) warn that TWFE with a heterogeneous continuous treatment can be badly misleading if this assumption fails.

What we did. We added a quadratic term $(\text{treat}_i \cdot \text{post}_i)^2$, and separately replaced the continuous treatment with tercile-by-post dummies (low/mid/high tariff shock). Quadratic fit: $\beta_{\text{lin}} = -1.49$ (SE 1.29), $\beta_{\text{quad}} = +6.06$ (SE 7.17). Tercile fit: high-shock $\times$ post = -0.073 (SE 0.055); mid-shock $\times$ post = -0.021 (SE 0.050).

Result. Neither the quadratic nor the tercile dummies are individually significant. The tercile specification is particularly revealing: the high-shock industries show only a tiny (and insignificant) additional decline relative to the low-shock baseline. There is no clear monotone dose-response.

4.9. HC1 robust SE as a Borusyak-Hull-Jaravel lower bound

Concern. Borusyak-Hull-Jaravel (2022) argue that with shift-share exposure the "correct" SE accounts for cross-industry correlation in the shock. An HC1 heteroskedasticity-robust SE is a conservative *lower bound* on the appropriate exposure-robust SE.

Result. HC1 $SE_{\text{full}} = 0.104$ (vs. clustered 0.436); HC1 $SE_{\text{no232}} = 0.116$ (vs. clustered 0.477). The HC1 SE is much smaller than the clustered SE, implying the clustered SE already imposes substantial skepticism. The exposure-robust SE sits between the two, so we treat the clustered-SE inference as conservative and consistent with the WCB and RI p-values above.

4.10. Laspeyres pre-period-basket tariff exposure

Concern. If the basket of goods inside an industry shifts after tariffs bite (e.g. importers substitute away from high-duty HTS codes), the post-period effective tariff rate used to compute the shock is partly endogenous. A Laspeyres-style measure using the pre-period basket weights would be immune to this bias.

What we did. We verified that our baseline_tariff variable is already constructed from the January-2025 (pre-shock) basket and the shock is $\Delta T_i = T_i^{\{\text{post}\}} - T_i^{\{\text{pre}\}}$. Refitting with this Laspeyres-equivalent shock reproduces the baseline exactly, as expected -- a useful sanity check that our treatment is already bias-free in this dimension.

4.11. Putting it all together

check	beta	se	p
Baseline DiD (full)	-0.494	0.436	0.273
Baseline DiD (no Sec. 232)	-1.044	0.477	0.046
Wild cluster bootstrap (full, B=1999)	-0.494	0.436	0.267

Wild cluster bootstrap (no 232, B=1999)	-1.044	0.477	0.057
Randomization inference (full, P=999)	-0.494	0.436	0.385
Randomization inference (no 232, P=999)	-1.044	0.477	0.163
Placebo date 2021-02	-0.43	0.388	0.283
Placebo date 2022-02	-0.428	0.379	0.274
Placebo date 2023-02	-0.452	0.38	0.251
Placebo date 2024-02	-0.47	0.381	0.234
HC1 robust SE (full)	-0.494	0.104	nan
HC1 robust SE (no 232)	-1.044	0.116	nan
PPML (full)	-0.151	0.269	0.576
PPML (no Sec. 232)	-0.622	0.469	0.184
Cross-industry slope (18 betas)	-0.494	0.449	0.272
Ibragimov-Muller one-sample t	nan	nan	0.089
Quadratic term (treat_intensity^2)	6.057	7.166	nan
Tercile: high-shock x post	-0.073	0.055	nan
Tercile: mid-shock x post	-0.021	0.05	nan

Table 2. Hardened inference results -- all 19 rows in one place. Numbers in parentheses are 95% randomization-inference quantiles for context.

5. Discussion

Stepping back, the picture that emerges after hardening is as follows. The full-sample point estimate is small and statistically indistinguishable from zero under every inference procedure we tried: clustered t, wild cluster bootstrap, Fisher randomization, PPML, Ibragimov-Müller. The no-Sec-232 sample point estimate is larger and survives clustered-t and the wild cluster bootstrap at the 10% level, but not at 5%, and does not survive Fisher randomization. Placebo-date tests reveal a pre-existing differential trend of roughly the same magnitude as the real estimate, implying that an honest reader should subtract most of the point estimate off the top. Finally, Rotemberg decomposition shows that our aggregate β is not even sign-consistent across the industries with the largest weights.

Three points deserve emphasis. First, the original significant result was real in the sense that clustered-t $p = 0.046$ in the no-232 sample -- but the procedure's nominal size is not the same as its actual size with $G = 18$. Second, the robust conclusion is not that tariffs had no effect; rather, fourteen months is too short a post-window and 18 clusters too small a sample to detect the effect even if it is of the order the no-232 point estimate suggests (-10% to a ten-point tariff increase). Third, the study documents a useful methodological workflow: report the naive estimate, then force the data to defend itself against every major criticism a modern applied econometrician would raise.

6. Conclusion

Using eighteen 3-digit NAICS manufacturing sub-sectors and USITC-calculated industry-level effective tariff rates, we estimate the employment effect of the February 2025 tariff package over its first fourteen post-months. Cluster-t inference returns a significant negative effect in the sub-sample that excludes Section-232-shielded industries. A comprehensive statistical hardening pass -- wild cluster bootstrap, Fisher randomization, leave-one-out, placebo dates, Rotemberg decomposition, PPML, Ibragimov-Müller, quadratic and tercile functional-form checks, and Borusyak-Hull-Jaravel bounds -- substantially weakens this finding. Our honest conclusion is that the 2025 tariffs' first-year employment footprint, as identified by this design and this data window, is small, noisy, and not robustly distinguishable from zero. The paper's larger contribution is methodological: a replicable workflow for stress-testing a TWFE DiD with a continuous shift-share treatment and few clusters.

Data and code

All code is at github.com/ash-grossman/ECO4370-Project. The reproducibility script `run_all.R` executes data ingestion, baseline DiD, 2SLS, the full robustness suite (`src/robustness.R`), and figures. Python companion scripts under `outputs/work/` produce the same hardening results independently. Employment data come from the FRED CES tables (series CES3231100001-CES3133700001); tariff data come from USITC DataWeb's Calculated Duties and Customs Value fields, aggregated from HTS-8 to NAICS-3 via the U.S. Census Bureau concordance. An `.Renv` file with a free FRED API key is required to re-ingest employment data and is excluded from the git repository.

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